# RGI *bwt*: A software pipeline for predicting AMR genes from raw bacterial short-read sequences

Michael G. DeGroote Institute for Infectious Disease Research

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**Amogelang R. Raphenya[1],** Brian P. Alcock[1], Finlay Maguire[2] & Andrew G. McArthur[1]

Michael G. DeGroote Institute for Infectious Disease Research, McMaster University, Hamilton, Canada[1]

Department of Community Health and Epidemiology, Dalhousie University, Halifax, Canada[2]

**Amos. R. Raphenya**

**Lead Developer I CARD I McArthur Lab**

## BACKGROUND

- Antibiotic resistance genes (ARGs) are a major One Health threat affecting humans, animals, and the environment
- Next-generation sequencing (NGS) enables detection and tracking of ARGs in diverse samples, including wastewater
- Large, unannotated sequencing datasets (1.3–10 GB) are increasingly common across sectors
- Current bioinformatics methods are slow, delaying identification of emerging ARGs threats
- Fast and accurate analytical tools are needed to address this gap
- The Resistance Gene Identifier (RGI) was developed to predict ARG determinants using the Comprehensive Antibiotic Resistance Database (CARD)

## RGI BWT PIPELINE

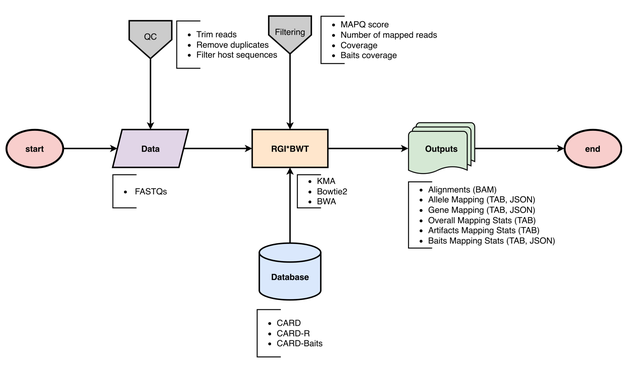


**Figure 1. RGI *bwt* pipeline.** Alignment software is selected (e.g., KMA) by users to align high quality short-read sequences to reference databases (CARD or CARD-R) for ARGs annotations. The pipeline produces multiple summary tabular files.

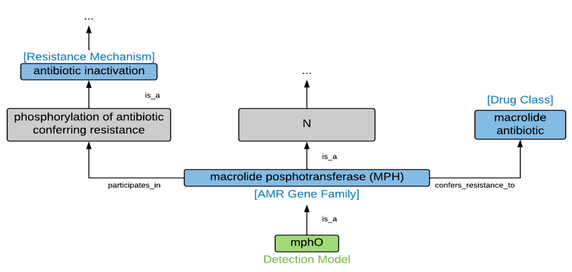


**Figure 2. The CARD Canonical reference sequences.** The sequences are manually curated from literature by AMR experts into an Antibiotic Resistance Ontology (ARO) structure, describing key relationships such as "Drug Class", "AMR Gene Family", and "Mechanism". The ARO and molecular data is then used to annotate predicted ARGs.

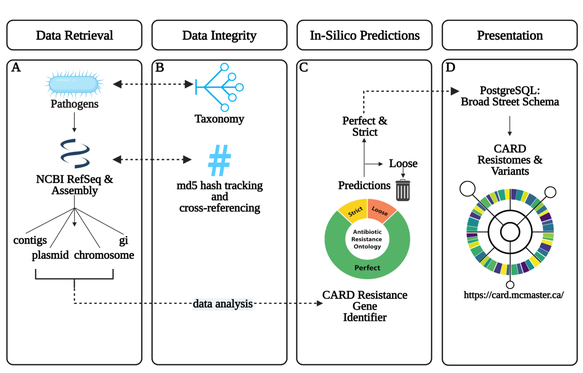


**Figure 3. The CARD-R dataset.** RGI is used to predict ARGs in NCBIs RefSeq and calculate the prevalence of each ARG curated in CARD. These data captures 279120 variants from 414 pathogens, 24291 chromosomes, 2662 genomic islands, 48212 plasmids, and 172216 WGS assemblies.

## WGS NEEDS MORE CARD-R REFERENCES THAN TARGETED CAPTURE

**Figure 4. Mapping to CARD Canonical vs CARD-R dataset.** (A) Reads mapped to whole genome sequencing (WGS) and target capture samples (17 samples in each group). The samples are from clinical pathogens in Hamilton Ontario. (B) The wastewater samples mapped to CARD-R and canonical. The two wasterwater samples were sourced from city of Hamilton Wastewater treatment plant. (AC = enriched with allCARD; CC = enriched with clinicalCARD, AO=enriched with Arbor-synthesized CARD v1 baits; HO = enriched with in-house-synthesized CARD v1 baits; SG=shotgun). More CARD-R reference sequences are needed to map reads for WGS as compared to targeted capture.

## RGI vs AMR++ WORKFLOW

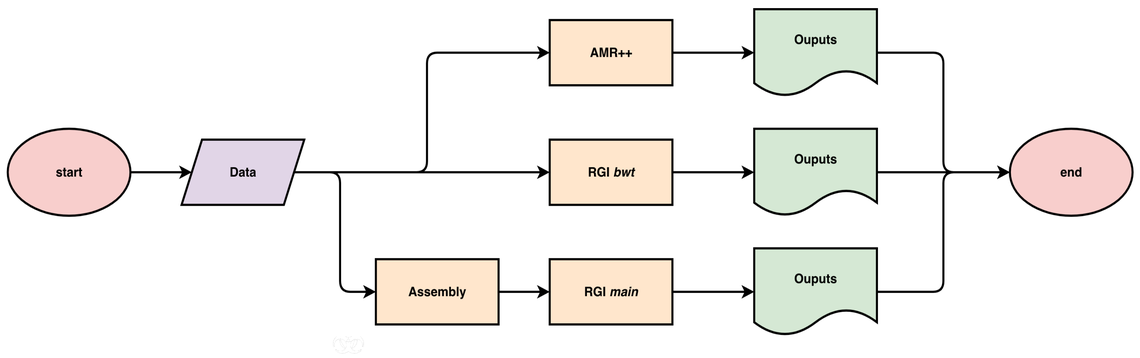


**Figure 5. RGI *bwt* vs AMR++ .** The figure shows the workflow used to compare results from RGI *bwt* and AMR++ starting with raw short-read sequences. The reads were assembled and only predicted "Perfect" annotations (RGI *main*) were used to compare AMR++ and RGI *bwt* results.

## RGI *bwt* PREDICTS MORE AMR GENES THAN AMR++

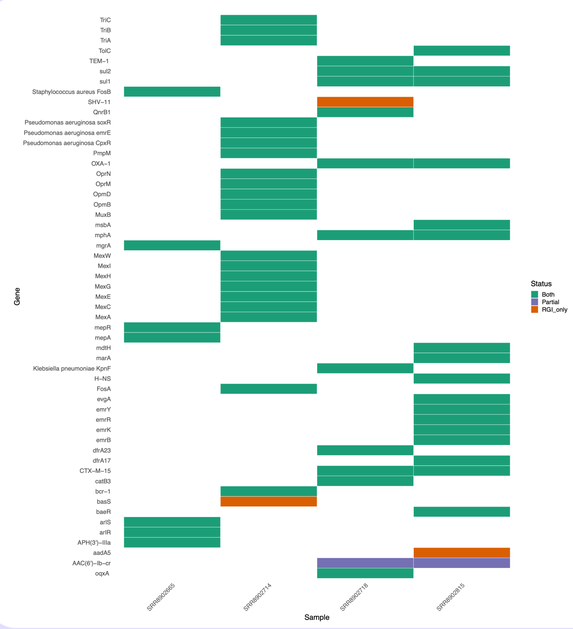


**Figure 6. Comparing annotated ARGs for both AMR++ and RGI *bwt*.** The 4 samples were assembled and perfect annotation were obtained (using RGI *main*) which are shown on the y-axis. Annotations from AMR++ are missing *blaSHV-11* from *Klebsiella pneumoniae* (SRR8902718), *basS* from *Pseudomonas aeruginosa* (SRR8902714), and *aadA5* from *Escherichia coli* (SRR8902815). RGI *bwt* annotations predicted all ARGs from the assembled genome isolates (RGI *main* step). "Both" indicates gene presence for RGI *bwt* and AMR++, "Partial" is prediction from same gene family, and "RGI_only" is ARGs predicted by RGI *bwt* only.

## CONCLUSIONS

- The RGI *bwt* provides an option to skip assembly, making it feasible to process large amounts of data to predict AMR determinants quickly
- Mining AMR genes from GenBank to produce the CARD-R dataset allows incorporation of non-clinical allelic variants as reference data
- RGI *bwt* is well-suited for AMR surveillance

## FUTURE DIRECTIONS

- Improve the RGI *bwt* pipeline to use GPUs
- Provide visualizations for the results
- Add taxonomic classification using tools such as minimap2 to complement CARD *k-mers*

## REFERENCES


Alcock BP, Huynh W, Chalil R, Smith K, and Raphenya A.R *et al.* Nucleic Acids Res. 51, D690–D699 (2023).

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## ACKNOWLEDGEMENTS


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